Aleksandr **Razin**

Sofia · Bulgaria

razin.x.aleks@gmail.com | razinaleksandr.github.io | RazinAleksandr | a-razin | Google Scholar

Research Scientist at INSAIT and ELLIS PhD student with ∼5 years of industry experience in applied research (Huawei) and production ML engineering. First & corresponding author of an ECCV 2026 paper on diffusion latent upscaling (#1 Hugging Face Daily Paper); led quantized post-training of night-HDR video restoration shipped on Huawei devices; built a multi-agent multimodal AI copilot for 2M+ monthly users; founded a computer-vision startup (acquired 2025). Research interests: **visual agents**, **generative world models**, **embodied AI**. M.Sc. ML (ITMO).

Experience

INSAIT

Research Scientist — *Sep 2026 – Present*

- Research on **visual agents** and **generative world models** for embodied AI with Dr. Jinjin Gu (INSAIT) and Prof. Marc Pollefeys (ETH Zurich)

Independent Researcher — *Dec 2025 – Sep 2026*

- First & corresponding author of **Latent Upscaling Adapter** (**ECCV 2026**, **#1 HF Daily Paper**); led theory and experiment design

TradingView

Senior ML Engineer — *Sep 2025 – Oct 2026*

- Engineered the harness and full offline/online evaluation for an agentic **AI copilot** (thousands of tools, hundreds of agents) for real-time chart analysis & modification; **2M+ monthly users**
- Built production **multimodal** search (text, charts/images, video) on fine-tuned embeddings + hybrid retrieval; ∼**80% precision** per modality
- Cut existing LLM-solution costs ∼**20%/month** by blending third-party APIs with local inference

Startup

Founder, Lead ML Engineer — *Apr 2025 – Sep 2025*

- Founded and hired a **5-engineer** product team; secured ∼€100K; **acquired (2025)** by a large IT company
- Shipped a retail **video-analytics** CV suite (detection, segmentation & tracking) to 3 of top-5 national retailers, saving ∼**€20K+/month**
- Post-trained **text-to-image / image-to-image** ad-generation models, improving identity preservation and lifting client ad **CTR** ∼**15%**

Huawei Research Institute

Research Scientist — *Dec 2023 – Apr 2025*

- Redesigned **video restoration** architectures (attention + convolution optimization), improving perceptual quality, detail reconstruction & color
- Optimized **diffusion training** (checkpoint-merging & loss design) for high-frequency texture synthesis; ∼**10%** gains over baselines (FID/PSNR)
- Led **quantized post-training** for the **Mate 70** night-HDR video restoration network (enhancement, super-resolution, joint denoising & demosaicing); refined strategy cut ghosting & flicker, shipping weekly to production

Research Intern — *May 2023 – Sep 2023*

- Improved domain **LLM** generation $> 30\%$ **ROUGE** via classification-agent query routing and LoRA + DPO post-training
- Enabled **multimodal** LLM agents by training metric-learned visual encoders on 100K+ sample datasets for visual question answering

ITMO University

Deep Learning Engineer — *Jun 2022 – May 2023*

- Built an end-to-end **data pipeline** (extraction, labeling, processing, analysis), raising detection & segmentation Precision/Recall
- Distilled banknote segmentation models (**20% fewer FLOPs**, no accuracy loss); **contrastive learning** lifted fake-stamp detection +10% F1

Mining University

Research Intern — *Jan 2021 – Dec 2021*

- Built **on-device** detection & segmentation models for obstacle & defect inspection by **pipeline robots** at oil & gas plants, saving ∼**€56K+/year**

Education

INSAIT, Sofia University – ETH, Zurich

PhD Computer Science, ELLIS PhD Program — *2026 – Present*

Advisors: Dr. Jinjin Gu (INSAIT), Prof. Marc Pollefeys (ETH)

ITMO University

M.Sc. Machine Learning (Honors), GPA: 4.9/5 – Thesis: *Enhancing Image Super-Resolution through Depth Estimation* — *2022 – 2024*

Teaching Assistant: *Advanced Machine Learning Technologies*

Mining University

B.Sc. Engineering, GPA: 4.9/5 – Thesis: *Real-time obstacle detection and analysis for robotic exploration with LiDAR* — *2018 – 2022*

Teaching Assistant: *Experimental Physics* · Mathematics Tutor: *Olympiad mathematics & programming*

Skills

Languages Python, Go, JAX
Tools Docker, Kubernetes, Kafka, AWS, Git, CI/CD, CUDA Toolkit, MLflow, Airflow, Kubeflow
Frameworks PyTorch, TorchVision, OpenCV, Transformers, Diffusers, DeepSpeed, DiffSynth-Studio, ComfyUI, vLLM, Google ADK
Concepts Vision / Language models, distributed train, post-train, inference & arch optimization, research, SWE
Language Skills English (fluent), Russian (native)